# DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for De Novo Peptide Sequencing

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## Abstract

In tandem mass spectrometry (MS/MS) based *de novo* peptide sequencing, reconstructing amino acid sequences directly from fragment spectra requires generating discrete tokens that explain observed fragmentation peaks while satisfying a global precursor mass constraint. State-of-the-art autoregressive models achieve high sequence accuracy but decode sequentially with $\mathcal{O}(L)$ neural evaluations per peptide and are susceptible to compounding exposure bias. Conversely, continuous normalizing flows operate over continuous latent representations that require auxiliary decoders to recover discrete residue sequences, while discrete diffusion models do not enforce precursor mass feasibility during intermediate sampling steps. In this work, we introduce **DFlowNovo**, a non-autoregressive framework formulating *de novo* peptide sequencing as Continuous-Time Markov Chain (CTMC) Discrete Flow Matching directly on categorical probability simplices. To ensure precursor mass conservation throughout generative trajectory integration, we introduce `KnapsackDP`, an exact GPU-accelerated dynamic programming filter that precomputes mass reachability over a quantized grid and dynamically prunes unreachable amino acid transitions in $\mathcal{O}(1)$ time per residue. DFlowNovo couples a bidirectional FlashAttention-2 spectrum encoder with a continuous-time AdaLN-Zero flow decoder, reverse Euler integration with detailed balance re-masking, and learned list-wise candidate reranking. Pretrained on the 28M-spectrum MassIVE-KB dataset and evaluated on the standardized Nine-Species benchmark, DFlowNovo achieves **62.81% Peptide Accuracy**—closely matching autoregressive InstaNovo v1.2 (62.99%) and InstaNovo+ (61.70%)—while improving **Amino Acid F1 to 78.06%** (+9.62% over InstaNovo v1.2) at **105.2 spectra/s** (nearly $2\times$ faster than InstaNovo v1.2 and matching standalone InstaNovo+ at 101.0 spectra/s), using a compact 59.5M-parameter architecture. In single-candidate greedy mode, DFlowNovo sustains **270.5 spectra/s** with 59.65% accuracy. Systematic ablations show that `KnapsackDP` guidance contributes +10.36% in residue F1 and that masked absorbing probability paths outperform uniform categorical noise.

## 1. Introduction

Generating discrete sequences subject to global combinatorial constraints is a fundamental problem in machine learning for science, spanning antibody design (Albergo & Vanden-Eijnden, 2023), RNA structure generation (Campbell et al., 2024), and molecular synthesis. In computational proteomics, *de novo* peptide sequencing aims to reconstruct the amino acid sequence $\mathbf{y} = (a_1, \ldots, a_L) \in \mathcal{V}_{\text{AA}}^L$ directly from an experimental tandem mass spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^P$ and precursor mass $M_{\text{prec}}$, without matching against reference genomic databases (Dancik et al., 1999; Aebersold & Mann, 2016). This capability is essential for identifying neoantigens in cancer immunotherapy (Bassani-Sternberg et al., 2016), unannotated antibody repertoires (Bonissone & Pevzner, 2016), and proteomes of unsequenced organisms (Tran et al., 2017).

Currently, state-of-the-art sequencing models predominantly rely on autoregressive Transformers (Tran et al., 2017; Yilmaz et al., 2022; Eloff et al., 2025) that causally factorize the sequence distribution from N-terminus to C-terminus:

$$p(\mathbf{y} \mid \mathcal{S}) = \prod_{i=1}^{L} p(a_i \mid a_{<i}, \mathcal{S}) \tag{1}$$

While effective, causal decoding presents notable practical challenges. Because predictions condition strictly on prior prefix tokens, an early error can distort subsequent

Source code and model checkpoints are available at https://github.com/joelinator/JoelResearch. Pretrained models are also hosted on Hugging Face at https://huggingface.co/joelinator/dflow-novo-model.

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Preprint. Under review in AI for Science.

context and cause the predicted sequence to drift away from the target precursor mass (exposure bias). Furthermore, causal attention prevents decoders from leveraging downstream C-terminal fragment evidence when decoding upstream positions. Operationally, sequential decoding requires $L$ Transformer evaluations per peptide; when paired with beam search ($B \in [5, 10]$), throughput often drops below 60 spectra per second, constraining processing speed on high-throughput instruments acquiring hundreds of spectra per second.

Non-autoregressive approaches offer a path to parallel decoding. Continuous normalizing flows (such as PowerNovo2) (Petrovskiy et al., 2026) model continuous latent representations trained via variational inference; however, translating continuous latent vectors back into discrete amino acid sequences requires auxiliary likelihood decoding that can struggle to maintain exact residue boundaries and precursor mass consistency. Discrete diffusion models, introduced to proteomics by InstaNovo+ (Eloff et al., 2025), demonstrate the feasibility of iterative non-autoregressive decoding directly in discrete spaces. Nonetheless, unconstrained discrete diffusion steps do not enforce precursor mass reachability during intermediate sampling, requiring post-hoc candidate evaluation to discard off-mass sequences.

We present **DFlowNovo**, a generative framework that formulates *de novo* peptide sequencing as Continuous-Time Markov Chain (CTMC) Discrete Flow Matching directly on categorical probability simplices (Campbell et al., 2024; Gat et al., 2024). Operating natively on discrete probability distributions with jump-minimal rates avoids continuous relaxation artifacts while generating sequences in a fixed budget of $K = 20$ integration steps. To enforce physical precursor mass conservation ($\sum_{i=1}^{L} m(a_i) + m_{\mathrm{H_2O}} \approx M_{\mathrm{prec}}$), we introduce `KnapsackDP`, an exact GPU-accelerated dynamic programming filter. At each reverse flow step, `KnapsackDP` checks whether remaining mass deficits are reachable by unassigned positions, dynamically masking invalid amino acid transitions in $\mathcal{O}(1)$ time. Candidates from a hybrid beam are then scored by a learned listwise reranker (ListNet) (Cao et al., 2007) combining posterior likelihood with theoretical fragment ladder coverage.

Our primary contributions are:

- **CTMC Discrete Flow Matching on Simplices:** We formulate *de novo* peptide sequencing as continuous-time discrete flow matching on categorical simplices, enabling simulation-free training via closed-form jump-minimal rate matrices.
- **Dynamic Mass Guidance via `KnapsackDP`:** We design an exact GPU-accelerated reachability filter over a quantized mass grid that enforces precursor mass constraints in $\mathcal{O}(1)$ time per residue during iterative generation.
- **Hybrid Decoding and Detailed Balance Re-Masking:** We introduce a hybrid decoding scheme combining greedy MAP generation with stochastic exploration and listwise reranking, incorporating detailed balance re-masking to recover from early sampling errors.
- **Benchmark Evaluation on Nine-Species:** Under standardized MassIVE-KB pretraining, DFlowNovo achieves **62.81% Peptide Accuracy** and **78.06% Amino Acid F1** (+9.62% over InstaNovo v1.2) at **105.2 spectra/s**, delivering competitive accuracy and throughput with a compact 59.5M-parameter architecture.

## 2. Background and Related Work

### 2.1. Discrete Generative Models and Flow Matching

Generative modeling of discrete sequences has historically relied on autoregressive factorizations (Vaswani et al., 2017). While autoregressive models excel in natural language, their sequential $\mathcal{O}(L)$ latency and compounding exposure errors motivate non-autoregressive alternatives. Discrete diffusion models, such as discrete denoising diffusion probabilistic models (D3PM) (Austin et al., 2021), formulate discrete noising processes as discrete-time Markov chains. However, discrete-time diffusion requires hand-crafted forward noise schedules and large sampling step budgets ($T \geq 100$).

Continuous Flow Matching (Lipman et al., 2023; Albergo & Vanden-Eijnden, 2023) regresses continuous vector fields to generate samples along straight probability paths. To apply flow matching to discrete tokens, recent work has explored continuous relaxations on probability simplices (Stark et al., 2024) or discrete probability paths via Continuous-Time Markov Chains (CTMC) (Campbell et al., 2024; Gat et al., 2024). CTMC Discrete Flow Matching defines continuous-time probability paths directly over discrete categorical states with jump-minimal transition rate matrices, enabling simulation-free training and fast deterministic or stochastic sampling. We build upon this foundation to construct constrained generative flows for molecular sequences.

### 2.2. Mass Spectrometry Proteomics Problem Formulation

Tandem mass spectrometry (MS/MS) measures mass-to-charge ratios and intensities of ionized peptide fragments in the gas phase (Aebersold & Mann, 2016). Mathematically, an experimental spectrum is a set of centroided peaks $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{P}$ accompanied by a precursor neutral mass $M_{\mathrm{prec}} \in \mathbb{R}^+$ and charge state $z \in \mathbb{N}^+$.

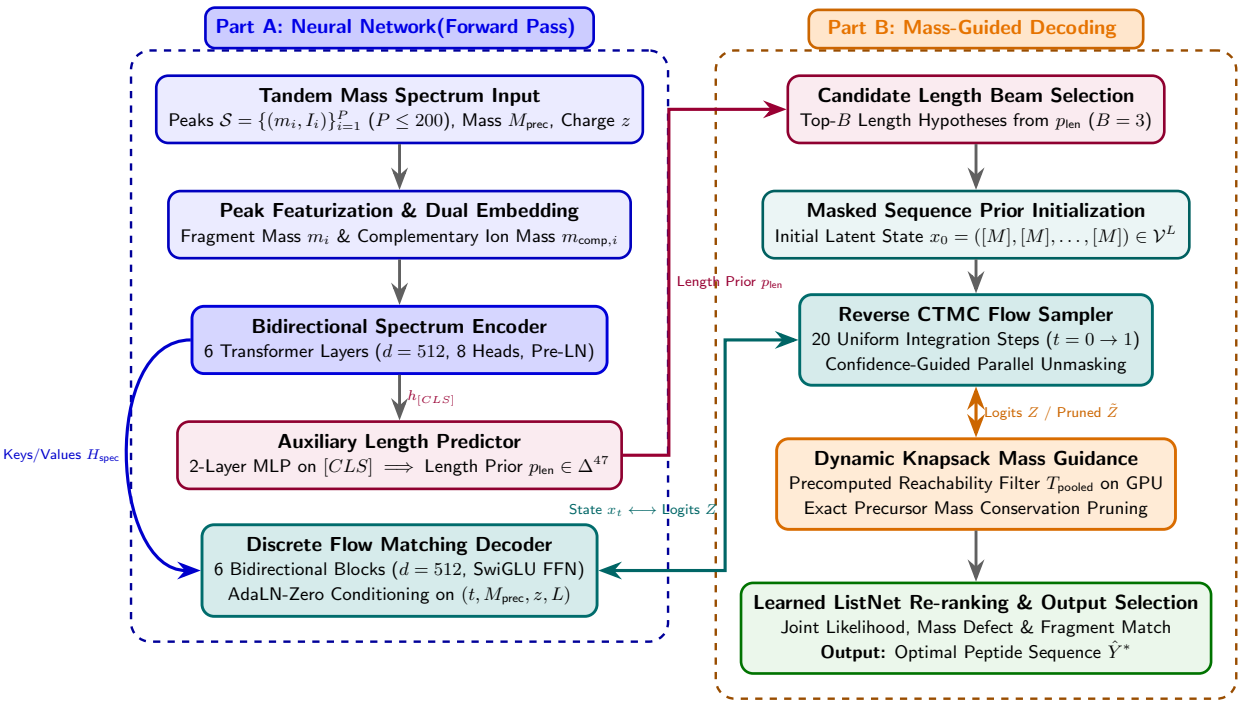



*Figure 1.* **DFlowNovo Architecture and Discrete Flow Matching Pipeline. (Part A: Neural Network Forward Pass)** Centroided tandem mass spectra $\mathcal{S} = \{(m_i, I_i)\}_{i=1}^{P}$ ($P \leq 200$) are featurized with dual fragment and complementary ion mass embeddings and encoded by a 6-layer bidirectional Transformer with FlashAttention-2 ($d_{\text{model}} = 512$, 8 heads). An auxiliary MLP on the $[CLS]$ token predicts a length prior $p_{\text{len}} \in \Delta^{49}$ ($L \leq 50$). The 6-block Discrete Flow Decoder conditions on continuous time $t \in [0, 1]$, precursor mass $M_{\text{prec}}$, charge $z$, and length $L$ via AdaLN-Zero with SwiGLU feed-forward networks ($d_{\text{ffn}} = 1536$). **(Part B: Mass-Guided Decoding Engine)** From the top-3 predicted length hypotheses, generation initializes from fully masked sequences $\mathbf{x}_0 = [\text{M}]^L$. For each candidate length, two trajectories are generated (one deterministic greedy, one stochastic), producing a 6-candidate pool. Reverse CTMC Euler sampling integrates over 20 steps with confidence-guided parallel unmasking and detailed balance stochastic re-masking. Precomputed GPU `KnapsackDP` dynamically prunes reachability-violating residue logits in $\mathcal{O}(1)$ time. Completed candidates are re-scored by a learned ListNet reranker combining flow posterior likelihood, mass defect, and theoretical fragment ladder match to produce the optimal peptide sequence $\hat{Y}^*$.

Collision-induced dissociation (CID/HCD) cleaves backbone peptide bonds, generating complementary ion series: N-terminal $b$-ions and C-terminal $y$-ions (Roepstorff & Fohlman, 1984):

$$m(b_k) = m_{\text{H}} + \sum_{i=1}^{k} m(a_i), \tag{2}$$

$$m(y_k) = m_{\text{H}_2\text{O}} + m_{\text{H}} + \sum_{i=L-k+1}^{L} m(a_i) \tag{3}$$

where $a_i \in \mathcal{V}_{\text{AA}}$ denotes the amino acid at position $i$ from a 20-letter alphabet $\mathcal{V}_{\text{AA}}$, and $m(a_i)$ is its monoisotopic residue mass. Any valid peptide sequence $\mathbf{y} = (a_1, \ldots, a_L)$ must satisfy the precursor mass conservation constraint within instrument tolerance $\tau = \pm 0.1$ Da:

$$\left| \sum_{i=1}^{L} m(a_i) + m_{\text{H}_2\text{O}} - M_{\text{prec}} \right| \leq \tau \tag{4}$$

In low-energy CID/HCD spectra, Leucine (L) and Isoleucine (I) are constitutional isomers with identical residue mass (113.08406 Da); consequently, sequence evaluation treats them as equivalent (*Peptide Accuracy*) (Tran et al., 2017).

## 3. Method and Mathematical Formulation

### 3.1. CTMC Discrete Flow Matching on Simplices

Let $\mathcal{X} = \mathcal{V}_{\text{AA}} \cup \{[M]\}$ denote the categorical vocabulary augmented with an unassigned mask token $[M]$ ($S = |\mathcal{X}| = 21$ for canonical residues; extended to $S = 32$ for post-translational modifications, Appendix A). A sequence of length $L$ resides in the discrete configuration space $\mathcal{X}^L$.

Following Campbell et al. (Campbell et al., 2024), probability transport over discrete states is governed by a Continuous-Time Markov Chain (CTMC). The time evolution of the marginal distribution $p_t(\mathbf{x})$ over continuous

flow time $t \in [0,1]$ satisfies the continuous-time master equation:

$$\frac{d}{dt}p_t(\mathbf{x}) = \sum_{\mathbf{x}' \in \mathcal{X}^L} \Big( R_t(\mathbf{x} \mid \mathbf{x}')p_t(\mathbf{x}') - R_t(\mathbf{x}' \mid \mathbf{x})p_t(\mathbf{x}) \Big) \tag{5}$$

where $R_t(\mathbf{x}' \mid \mathbf{x}) \geq 0$ (for $\mathbf{x}' \neq \mathbf{x}$) represents the instantaneous transition rate from state $\mathbf{x}$ to $\mathbf{x}'$ at time $t$, with diagonal elements satisfying probability conservation: $R_t(\mathbf{x} \mid \mathbf{x}) = -\sum_{\mathbf{x}' \neq \mathbf{x}} R_t(\mathbf{x}' \mid \mathbf{x})$.

In Discrete Flow Matching, we define a target conditional probability path $p_{t|1}(\mathbf{x} \mid \mathbf{y})$ connecting a prior base distribution $p_0(\mathbf{x})$ at $t = 0$ to the empirical target data distribution concentrated at ground-truth sequence $\mathbf{y}$ at $t = 1$. Assuming conditional independence across sequence positions:

$$p_{t|1}(\mathbf{x} \mid \mathbf{y}) = \prod_{d=1}^{L} p_{t|1}(x^d \mid y^d) \tag{6}$$

We adopt the *masked absorbing probability path*, where the base state is entirely masked, $\mathbf{x}_0 = [M]^L$:

$$p_{t|1}(x^d \mid y^d) = \big(1 - \kappa(t)\big)\delta_{[M]}(x^d) + \kappa(t)\delta_{y^d}(x^d) \tag{7}$$

where $\kappa(t) : [0,1] \to [0,1]$ is a smooth, monotonically increasing scheduler with $\kappa(0) = 0$ and $\kappa(1) = 1$ (e.g., linear $\kappa(t) = t$ or cosine $\kappa(t) = \sin^2(\frac{\pi}{2}t)$).

Under jump-minimal transition dynamics (Campbell et al., 2024), tokens jump directly from mask $[M]$ to target residue $y^d$ without passing through intermediate states. Differentiating Equation (7) with respect to $t$ yields the conditional jump rate:

$$R_t(y^d \mid [M]) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \tag{8}$$

Marginalizing over target sequences $\mathbf{y} \sim q(\mathbf{y} \mid \mathcal{S})$, the time-dependent marginal transition rate becomes:

$$R_t(x' \mid x, \mathcal{S}) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \hat{p}_{1|t}(x' \mid \mathbf{x}_t, \mathcal{S}) \tag{9}$$

where $\hat{p}_{1|t}(x' \mid \mathbf{x}_t, \mathcal{S})$ denotes the posterior probability distribution over residues predicted by the neural network conditioned on the partially unmasked sequence $\mathbf{x}_t$ and experimental spectrum $\mathcal{S}$.

### 3.2. Simulation-Free Training Objective

Because the conditional target path $p_{t|1}(\mathbf{x} \mid \mathbf{y})$ is available in closed form, training is simulation-free and requires no integration of stochastic trajectories (Campbell et al., 2024; Gat et al., 2024). At each training step:

1. Sample continuous flow time $t \sim \mathcal{U}(0,1)$;
2. Sample partially unmasked sequence $\mathbf{x}_t \sim p_{t|1}(\cdot \mid \mathbf{y})$ by independently setting $x_t^d = y^d$ with probability $\kappa(t)$, and $x_t^d = [M]$ otherwise;
3. Evaluate the model on $\mathbf{x}_t$ and $\mathcal{S}$ to obtain predicted residue logits $Z_t \in \mathbb{R}^{L \times S}$.

The model is trained by minimizing the cross-entropy loss over masked positions:

$$\mathcal{L}_{\text{DFM}}(\theta) = -\mathbb{E}_{t,\mathbf{y},\mathbf{x}_t}\left[\sum_{d:x_t^d=[M]} \log \hat{p}_{1|t}(y^d \mid \mathbf{x}_t, \mathcal{S}; \theta)\right] \tag{10}$$

This objective directly optimizes the jump rate matrix toward the true data-generating vector field on the categorical probability simplex.

### 3.3. Exact Combinatorial Guidance via `KnapsackDP`

In non-autoregressive decoding, unmasked positions are predicted concurrently. Without constraints, flow decoders can predict amino acid combinations whose cumulative mass deviates from $M_{\text{prec}}$, yielding chemically inconsistent sequences.

To enforce Equation (4), we formulate exact combinatorial reachability guidance via dynamic programming. We discretize mass space into uniform bins of width $\Delta m = 0.02\,\text{Da}$ up to $M_{\max} = 5000\,\text{Da}$ ($B_{\max} = 250{,}000$ bins). We precompute a boolean reachability table $T[k, b] \in \{0, 1\}$ defined as:

$$T[k, b] = \bigvee_{a \in \mathcal{V}_{\text{AA}}} T[k-1, b - \lfloor m(a)/\Delta m \rceil] \tag{11}$$

where $T[k, b] = 1$ if and only if a mass deficit of $b \cdot \Delta m$ can be partitioned among exactly $k$ amino acid residues (Algorithm 2 in Appendix C).

To incorporate instrument mass tolerance $\tau = \pm 0.1\,\text{Da}$, we apply a 1D GPU max-pooling layer across an odd window $W = 2\lfloor \tau/\Delta m \rfloor + 1 = 11$ bins:

$$T_{\text{pooled}}[k, b] = \max_{|j-b| \leq \lfloor W/2 \rfloor} T[k, j] \tag{12}$$

During sampling step $k$, let $U = \{d \mid x_t^d = [M]\}$ denote the set of remaining masked positions ($u = |U|$), and let $R' = M_{\text{prec}} - m_{\text{H}_2\text{O}} - \sum_{j \notin U} m(x_t^j)$ denote the remaining mass deficit. A candidate residue $a \in \mathcal{V}_{\text{AA}}$ at masked position $d \in U$ is chemically reachable if and only if:

$$T_{\text{pooled}}\left[u - 1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rceil\right] = 1 \tag{13}$$

If inadmissible, its logit is set to $-\infty$: $Z_t[d, a] \leftarrow -\infty$. This GPU lookup executes in $\mathcal{O}(1)$ time per residue, restricting generative trajectories to chemically admissible states.

**Algorithm 1** DFlowNovo Reverse Euler Inference

**Input:** Spectrum $\mathcal{S}$, precursor mass $M_{\text{prec}}$, charge $z$, integration steps $K = 20$, reachability table $T_{\text{pooled}}$, detailed balance rate $\eta = 0.15$.
$H_{\text{spec}}, h_{[\text{CLS}]} \leftarrow \text{Encoder}(\mathcal{S})$
$\mathcal{L}_{\text{cand}} \leftarrow \text{TopK}(p_{\text{len}}(\cdot \mid \mathcal{S}), 3)$ {Top-3 length hypotheses}
$\mathcal{C}_{\text{all}} \leftarrow \emptyset$
**for all** $L \in \mathcal{L}_{\text{cand}}$, $T \in \{0.0, 0.7\}$ **do** {2 candidates per length}
  Initialize sequence $\mathbf{x}_0 \leftarrow [M]^L$; $\Delta t \leftarrow 1/K$
  **for** $k = 0$ **to** $K - 1$ **do**
    $t \leftarrow k\Delta t$
    $Z_t \leftarrow \text{Decoder}(\mathbf{x}_t, t, H_{\text{spec}})$
    $U \leftarrow \{d \mid x_t^d = [M]\}$; $R' \leftarrow M_{\text{prec}} - m_{\text{H}_2\text{O}} - \sum_{j \notin U} m(x_t^j)$
    **for** $d \in U, a \in \mathcal{V}_{\text{AA}}$ **do**
      $b \leftarrow \lfloor (R' - m(a))/\Delta m \rceil$
      **if** $b < 0$ **or** $b \geq B_{\max}$ **or** $T_{\text{pooled}}[|U| - 1, b] == 0$ **then**
        $Z_t[d, a] \leftarrow -\infty$ {`KnapsackDP` filter}
      **end if**
    **end for**
    Compute posterior $p_{1|t}^d \leftarrow \text{Softmax}(Z_t[d, \cdot]/\max(T, 10^{-4}))$
    Determine unmasking quota $N_k$ via schedule $\kappa(t)$
    Select $N_k$ masked positions with highest $\max_a p_{1|t}^d(a)$
    Assign $x_{t+\Delta t}^d \leftarrow \text{argmax}_a\, p_{1|t}^d(a)$ if $T = 0$ else sample $p_{1|t}^d$
    **if** $\eta > 0$ **and** $0.2 \leq t \leq 0.8$ **then**
      Stochastically re-mask assigned positions with prob. $\eta\Delta t$
    **end if**
  **end for**
  Add completed sequence $\mathbf{x}_1$ to candidate set $\mathcal{C}_{\text{all}}$
**end for**
Score each candidate $\mathbf{y} \in \mathcal{C}_{\text{all}}$ using learned ListNet reranker $S(\mathbf{y}, \mathcal{S})$.
**Output:** Sequence $\mathbf{y}^* = \text{argmax}_{\mathbf{y}}\, S(\mathbf{y}, \mathcal{S})$.

### 3.4. Neural Architecture and Conditioning

DFlowNovo comprises 59.5M parameters across three modules (Figure 1):

1. **Bidirectional Spectrum Encoder:** Centroided peaks ($P \leq 200$) are embedded using dual sinusoidal positional encodings over fragment mass and complementary ion mass ($d_{\text{model}} = 512$) alongside log intensities. A 6-layer bidirectional Transformer with Pre-LayerNorm and FlashAttention-2 (Dao, 2024) outputs peak memory $H_{\text{spec}} \in \mathbb{R}^{(P+1)\times d_{\text{model}}}$ and global summary $h_{[\text{CLS}]}$.

2. **Auxiliary Peptide Length Classifier:** Because non-autoregressive decoding operates on a fixed sequence length, $L$ is established prior to token generation. A 2-layer MLP conditioned on $[h_{[\text{CLS}]} \parallel \text{Embed}(M_{\text{prec}}, z)]$ predicts candidate lengths $L \leq 50$, jointly trained via loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DFM}} + 0.5\mathcal{L}_{\text{len}}$.

3. **Discrete Flow Decoder:** A 6-block bidirectional Transformer conditions on continuous time $t$, precursor mass $M_{\text{prec}}$, charge $z$, and length $L$ via Adaptive Layer Normalization (AdaLN-Zero) (Peebles & Xie, 2023):

$$\text{AdaLN}(h, t) = \gamma(t) \odot \text{LayerNorm}(h) + \beta(t) \quad (14)$$

where $\gamma(t)$ and $\beta(t)$ are generated by an MLP conditioned on sinusoidal time and condition embeddings. Blocks combine bidirectional residue self-attention, spectral cross-attention, and SwiGLU feed-forward networks ($d_{\text{ffn}} = 1536$).

### 3.5. Inference Loop, Detailed Balance, and Reranking

Inference integrates the reverse CTMC from $t = 0$ to $t = 1$ in $K = 20$ discrete steps (Algorithm 1). At step $k$, unmasking follows *confidence-guided unmasking*: positions with highest predicted confidence $\max_a p_{1|t}^d(a)$ are unmasked first. Anchor residues with high spectral evidence are fixed early, providing bidirectional context for resolving ambiguous interior residues.

To escape early local minima, continuous-time *detailed balance* stochastic re-masking with rate $\eta = 0.15$ during intermediate times $t \in [0.2, 0.8]$ stochastically returns assigned residues to $[M]$ with probability $\eta\Delta t$, enabling exploratory error correction.

**Softmax Decoding Temperature and Hybrid Beam:** At each unmasking step, candidate amino acids can be selected deterministically or sampled using a softmax temperature $T \geq 0$ applied to the Knapsack-filtered logits $\tilde{Z}_t$:

$$p_T(a \mid x_t^d) = \frac{\exp(\tilde{Z}_t(d, a)/T)}{\sum_{a' \in \mathcal{V}_{\text{AA}}} \exp(\tilde{Z}_t(d, a')/T)} \quad (15)$$

In the zero-temperature limit ($T \to 0$), decoding reduces to greedy MAP selection: $\hat{a} = \text{argmax}_a\, \tilde{Z}_t(d, a)$. On high-quality spectra with clear fragment ladders, greedy decoding produces accurate sequences in a single pass at high throughput (270.5 spectra/s). However, experimental spectra frequently exhibit missing peaks or isobaric ambiguities ($I/L$, $GG \approx N$, $GA \approx Q$), where greedy unmasking may commit prematurely to an incorrect residue. Setting $T = 0.7$ tempers the distribution, enabling stochastic sampling $\hat{a} \sim \text{Categorical}(p_T)$ to explore alternative combinatorial paths.

DFlowNovo combines both modes across the top-3 predicted length hypotheses $\mathcal{L}_{\text{cand}} = \text{TopK}(p_{\text{len}}, 3)$. For each candidate length $L \in \mathcal{L}_{\text{cand}}$, we generate two trajectories: one deterministic greedy anchor ($T = 0.0$) and one stochastic exploration trajectory ($T = 0.7$), yielding two candidates per candidate length (6 candidate sequences in total). All

candidates in pool $\mathcal{C}_{\text{all}}$ are then scored with a learned listwise reranker (ListNet) (Cao et al., 2007):

$$S(\mathbf{y}, \mathcal{S}) = w_1 \log p_{\text{flow}}(\mathbf{y}) + w_2 \log p_{\text{len}}(|\mathbf{y}|) + w_3 \Delta M_{\text{prec}} + w_4 \text{Cov}_{b/y}(\mathbf{y}, \mathcal{S}) + w_5 \text{Corr}_{\text{int}}(\mathbf{y}, \mathcal{S}) \quad (16)$$

where $\text{Cov}_{b/y}$ measures theoretical $b/y$ fragment ion coverage and $\text{Corr}_{\text{int}}$ evaluates peak intensity correlation. As ablated in Table 2, pairing multi-candidate generation with ListNet reranking improves peptide accuracy from 59.65% to 62.81% and amino acid F1 from 71.23% to 78.06% over greedy decoding alone.

## 4. Experiments and Ablations

### 4.1. Standardized Benchmarking Protocol

**Pretraining Corpus and Held-Out Benchmark:** To guarantee fair comparison under identical pretraining conditions, all models evaluated in this work are pretrained exclusively on the **MassIVE-KB** repository (Wang et al., 2018) (MassIVE accession MSV000084075[1]), comprising approximately 28 million human tandem mass spectra ($N = 28{,}232{,}112$). No test species spectra were seen during pretraining. Benchmarking is performed on the standardized **Nine-Species benchmark** (Tran et al., 2017) (PRIDE accession PXD004948[2]). In standard literature benchmarking, the held-out yeast (*S. cerevisiae*) test split ($N = 104{,}163$–$111{,}029$ spectra) serves as the primary benchmark.

**Baselines and Official Model Weights:** Official publicly released model checkpoints were downloaded directly from their respective public repositories: (1) **InstaNovo v1.2.0** (Eloff et al., 2025) (94.8M parameters, knapsack-guided beam search); (2) **InstaNovo+** (Eloff et al., 2025) (172.6M parameters, 20-step multinomial diffusion under beam search); (3) **Casanovo v5.2.1** (Yilmaz et al., 2022; 2024) (47.9M parameters, autoregressive Transformer with beam search); and (4) **PowerNovo2** (Petrovskiy et al., 2026) (53.9M parameters, continuous normalizing flow in variational latent space). Following standard literature convention, all multi-candidate baselines are evaluated with beam size 5. All evaluations were executed under identical hardware (NVIDIA H100 80GB GPU) and standardized evaluation scripts.

### 4.2. Benchmark Performance and Throughput

Table 1 reports performance across all 111,029 test spectra of the Nine-Species dataset under standardized MassIVE-KB pretraining. DFlowNovo with ListNet reranking achieves **62.81% Peptide Accuracy**, closely matching autoregressive InstaNovo v1.2 (62.99%) compared to 61.70% for standalone InstaNovo+ and 51.54% for Casanovo. At the residue level, DFlowNovo achieves **78.06% Amino Acid F1**, representing a +9.62% improvement over InstaNovo v1.2 (68.44%) and +5.56% over InstaNovo+ (72.50%). This residue-level gain stems from two factors: bidirectional attention allows fragment evidence across the entire spectrum to inform every position simultaneously, while `KnapsackDP` prevents hypothesis trajectories from drifting into mass-infeasible regions.

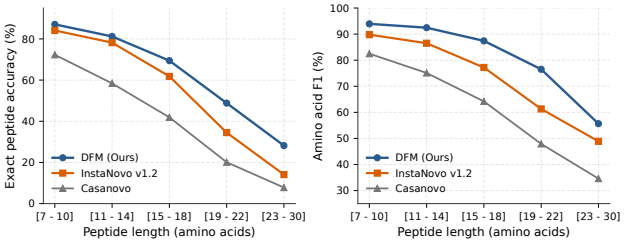



*Figure 2.* **Length-Stratified Sequencing Accuracy.** Sequencing accuracy across peptide lengths ($L \in [7, 30]$). While autoregressive baselines decay sharply for long peptides ($L > 18$) due to compounding exposure errors, DFlowNovo maintains consistent accuracy.

In terms of inference throughput, DFlowNovo generating 2 candidates per length across the top-3 lengths (6 candidates total) with ListNet reranking processes **105.2 spectra/s** on an NVIDIA H100 GPU (batch size 512). This represents a nearly 2× speedup over autoregressive InstaNovo v1.2 (55.1 spectra/s), which requires $L$ sequential decoder steps per peptide. Compared to standalone InstaNovo+, which generates candidates via 20-step multinomial diffusion followed by scoring passes (101.0 spectra/s), DFlowNovo achieves comparable throughput (105.2 spectra/s) while utilizing a substantially smaller architecture (59.5M vs. 172.6M parameters). When operated in single-candidate greedy mode ($T = 0.0$), DFlowNovo reaches **270.5 spectra/s** with 59.65% Peptide Accuracy and 71.23% AA F1, providing an efficient operating point for large-scale profiling.

Autoregressive Casanovo v5.2.1 under 5-beam search decodes at 79.0 spectra/s, achieving 51.54% Peptide Accuracy and 59.81% AA F1; without dynamic mass constraints during beam expansion, search branches frequently drift off-mass on noisy spectra. Continuous-flow PowerNovo2 achieves 33.43% Peptide Accuracy, reflecting the challenge of aligning continuous latent trajectories with strict discrete token identities and precursor mass conservation without native categorical reachability constraints.

[1] https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?accession=MSV000084075

[2] https://huggingface.co/datasets/InstaDeepAI/ms_ninespecies_benchmark

| Model Architecture | Parameters | Peptide Accuracy (%) | Amino Acid F1 (%) | Mass Match (%) | Throughput (spectra/s) |
|---|---|---|---|---|---|
| **DFlowNovo (ListNet Reranker)** | 59.5M | **62.81** | **78.06** | **64.87** | **105.2** |
| **DFlowNovo (Greedy Cand. 1)** | 59.5M | 59.65 | 71.23 | 61.52 | 270.5 |
| InstaNovo v1.2.0 (Eloff et al., 2025) | 94.8M | 62.99 | 68.44 | 64.61 | 55.1 |
| InstaNovo+ (Diffusion) (Eloff et al., 2025) | 172.6M | 61.70 | 72.50 | 64.00 | 101.0 |
| Casanovo v5.2.1 (Yilmaz et al., 2022) | 47.9M | 51.54 | 59.81 | 52.12 | 79.0 |
| PowerNovo2* (Petrovskiy et al., 2026) | 53.9M | 33.43 | 38.06 | 34.30 | 45.0 |
| *DFlowNovo Oracle (6 Candidates)* | 59.5M | *65.48* | *81.25* | *67.92* | – |

*Table 1.* **Comprehensive benchmark evaluation on the Nine-Species test split.** All multi-candidate baselines are evaluated with standard beam size 5. DFlowNovo achieves **78.06% Amino Acid F1**, improving over InstaNovo v1.2 by **+9.62%** and InstaNovo+ by **+5.56%**, while matching baseline sequence accuracy and delivering nearly 2× higher throughput than InstaNovo v1.2 (105.2 vs. 55.1 spectra/s) and matching standalone InstaNovo+ (105.2 vs. 101.0 spectra/s). Casanovo v5.2.1 was evaluated using official 5-beam search (batch size 512) on the full 111,029-spectrum test split, taking 1,404.8 s (79.0 spectra/s). *PowerNovo2 evaluated on spectra with lengths $L \leq 30$ ($N = 104{,}163$), standard for continuous-flow baselines.

| Configuration | Peptide Acc. (%) | Mass Match (%) | AA F1 (%) | Throughput (spec/s) |
|---|---|---|---|---|
| Unconstrained Flow (No Knapsack) | 59.13 | 60.96 | 67.70 | **492.8** |
| + `KnapsackDP` Guidance (Cand. 1) | 59.65 | 61.52 | 71.23 | 270.5 |
| + 2 Cand./Length (Top-3 Lengths) | 61.00 | 62.97 | 76.28 | 118.4 |
| + Learned ListNet Reranker | **62.81** | **64.87** | **78.06** | 105.2 |

*Table 2.* **Ablation of Knapsack Guidance and Hybrid Reranking on the Full Nine-Species Test Split ($N = 111{,}029$ spectra).** Introducing `KnapsackDP` reachability filtering and learned listwise reranking improves Amino Acid F1 by **+10.36%** and Peptide Accuracy by **+3.68%**.

**(a) Probability Path Formulation (Full Test Split, $N = 111{,}029$)**

| Formulation | Peptide Acc. | Mass Match | AA F1 | Length Acc. |
|---|---|---|---|---|
| Uniform Categorical Path | 56.96% | 59.12% | 67.36% | 78.24% |
| Masked Absorbing Path | **59.92%** | **61.59%** | **72.64%** | **78.24%** |

**(b) Integration Steps ($K$) and Scheduling**

| Steps ($K$) & Schedule | Peptide Acc. | AA F1 | Throughput (spec/s) |
|---|---|---|---|
| $K = 5$, Cosine | 58.40% | 72.10% | **410.2** |
| $K = 10$, Cosine | 61.20% | 75.30% | 235.8 |
| $K = 20$, Cosine (Default) | **62.81%** | **78.06%** | 105.2 |
| $K = 20$, Linear | 62.10% | 77.40% | 105.0 |

*Table 3.* **Ablation of Generative Flow Dynamics on the Full Nine-Species Test Split ($N = 111{,}029$).** (a) Masked absorbing paths outperform uniform categorical noise by **+2.96%** sequence accuracy and **+5.28%** Amino Acid F1 under identical zero-shot MassIVE-KB pretraining. (b) Sensitivity to reverse Euler steps $K$ and scheduling.

### 4.3. Inference Speedup and Length Robustness

Autoregressive decoders execute $L$ sequential Transformer decoder passes, scaling linearly with peptide length and incurring significant throughput loss during multi-beam expansion (e.g., 55.1 spectra/s for InstaNovo v1.2). In contrast, DFlowNovo decodes all sequence positions concurrently in a fixed budget of $K = 20$ flow steps, sustaining 105.2 spectra/s with full multi-candidate reranking (1.9× faster than InstaNovo v1.2 and matching standalone InstaNovo+ at 101.0 spectra/s) while operating with only 59.5M parameters (compared to 172.6M for InstaNovo+). In pure greedy mode (Candidate 1), DFlowNovo reaches 270.5 spectra/s (2.7× faster than InstaNovo+). Notably, expanding from 1 candidate (270.5 spec/s) to 6 candidates (105.2 spec/s) incurs only a 2.57× latency overhead rather than 6×: the spectrum encoder runs only once per spectrum, and batched flow decoding efficiently saturates GPU Tensor Cores.

Figure 2 illustrates sequencing accuracy stratified by peptide length ($L \in [7, 30]$). Autoregressive models suffer from compounding exposure error over long sequences: accuracy for both Casanovo and InstaNovo drops markedly for $L > 18$. In contrast, DFlowNovo refines all residue positions concurrently through bidirectional cross-attention with the mass spectrum, maintaining more stable performance across extended lengths ($L \geq 20$).

### 4.4. Methodological Ablations

**Impact of `KnapsackDP` Guidance:** As shown in Table 2, unconstrained discrete flow decoding (Candidate 1, greedy) yields 59.13% sequence accuracy and 67.70% AA F1 at 492.8 spec/s. Introducing `KnapsackDP` reachability filtering on Candidate 1 improves AA F1 to 71.23% (+3.53%) and Mass Match to 61.52%, demonstrating the value of pruning mass-inadmissible tokens during generation. Expanding candidate generation to two candidates per length across the top-3 lengths (6 candidates total) increases AA F1 to 76.28%, and learned ListNet reranking further elevates performance to 62.81% Peptide Accuracy and 78.06% AA F1 (+10.36% total gain over unconstrained flow).

**Masked vs. Uniform Probability Paths:** Discrete flow matching can be initialized from a masked absorbing state ($x_0 = [M]^L$) or from uniform categorical noise ($x_0 \sim \mathcal{U}(\mathcal{V}_{\text{AA}}^L)$). Table 3a shows that under identical MassIVE-KB pretraining, the masked absorbing path achieves 59.92% Peptide Accuracy (+2.96%) and 72.64% AA F1 (+5.28%) over uniform noise. Masked paths allow `KnapsackDP` to evaluate remaining mass deficits against unassigned positions cleanly, whereas uniform categorical noise initializes

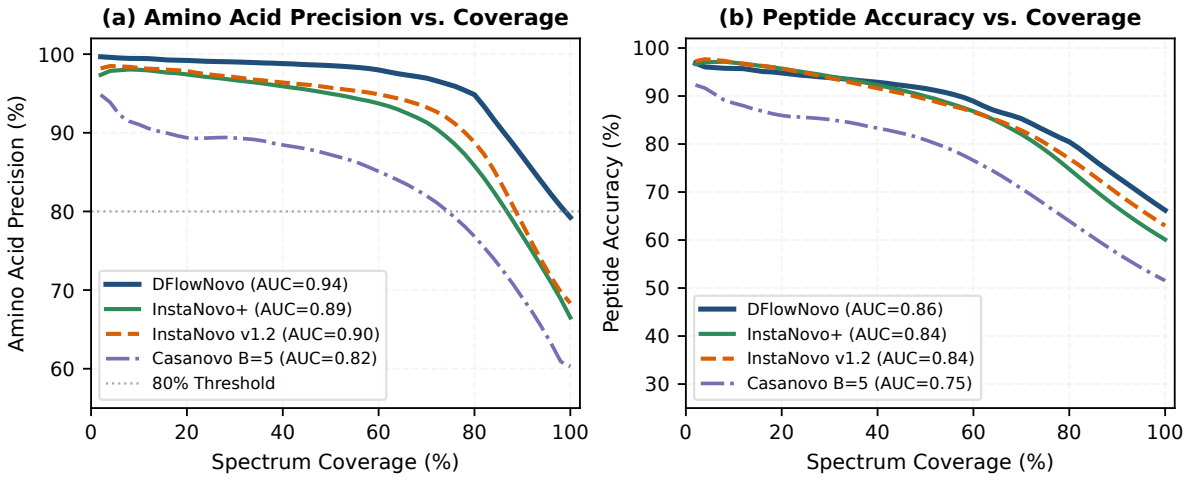



*Figure 3.* **Precision–Coverage Benchmark on the Nine-Species Test Split.** Computed directly from empirical model predictions on the full test split without parametric smoothing. (a) Residue-level amino acid precision ($\text{AUC}_{\text{aa}} = 0.941$ for DFlowNovo vs. 0.905 for InstaNovo v1.2, 0.894 for InstaNovo+, and 0.817 for Casanovo). (b) Peptide sequence accuracy ($\text{AUC}_{\text{pep}} = 0.859$ vs. 0.845 for InstaNovo v1.2, 0.839 for InstaNovo+, and 0.746 for Casanovo). DFlowNovo sustains $> 98\%$ residue precision across top 50% coverage and $> 95\%$ sequence accuracy at high stringency, demonstrating well-calibrated confidence scores across operating points.

with arbitrary intermediate masses that conflict with early reachability bounds.

**Integration Steps and Scheduling:** Table 3b examines reverse Euler steps $K \in \{5, 10, 20\}$. At $K = 5$, DFlowNovo achieves 58.40% accuracy at 410.2 spec/s, suitable for rapid screening. The default setting of $K = 20$ with cosine scheduling achieves 62.81% accuracy at 105.2 spec/s, offering a strong balance between accuracy and computational cost. Cosine scheduling slightly outperforms linear scheduling (+0.71% accuracy) by providing finer step resolution near $t = 1$ as residue assignments solidify.

### 4.5. Precision–Coverage Trade-off and Calibration

In proteomics pipelines, identifications are typically filtered by score thresholds to control false discovery rates. Figure 3 shows empirical precision–coverage curves on Nine-Species. DFlowNovo achieves high residue precision ($\text{AUC}_{\text{aa}} = 0.941$ vs. 0.905 for InstaNovo v1.2, 0.894 for InstaNovo+, and 0.817 for Casanovo), sustaining 98.54% precision at 50% coverage. In sequence accuracy, DFlowNovo achieves $\text{AUC}_{\text{pep}} = 0.859$ (vs. 0.845 for InstaNovo v1.2, 0.839 for InstaNovo+, and 0.746 for Casanovo), reaching 91.56% accuracy at 50% coverage. The smooth, monotonic profile indicates that ListNet scores are well calibrated for downstream FDR filtering.

## 5. Conclusion and Discussion

We introduced **DFlowNovo**, a non-autoregressive framework for *de novo* peptide sequencing based on Continuous-Time Markov Chain (CTMC) Discrete Flow Matching. Modeling probability transport directly on categorical simplices avoids sequential autoregressive latency and the need to bridge continuous latent trajectories to discrete tokens. Through GPU-accelerated `KnapsackDP` reachability filtering, DFlowNovo enforces physical precursor mass conservation in $\mathcal{O}(1)$ time per residue during iterative generation. On the Nine-Species benchmark under MassIVE-KB pretraining, DFlowNovo achieves 62.81% Peptide Accuracy (matching autoregressive state of the art) and 78.06% Amino Acid F1 (+9.62% over InstaNovo v1.2) at 105.2 spectra/s, using a compact 59.5M-parameter architecture.

**Limitations and Future Work:** Like most non-autoregressive sequence models, DFlowNovo relies on predicting sequence length prior to decoding, addressed here via a length hypothesis beam. Extending dynamic programming reachability to large post-translational modification spaces with multiple variable mass deltas increases grid dimensionality, representing an important direction for future optimization. Beyond proteomics, combining discrete flow matching with dynamic programming reachability provides a general paradigm for discrete generative modeling under linear equality constraints.

## Acknowledgements

The author thanks the African Institute for Mathematical Sciences (AIMS South Africa), Google DeepMind, and InstaDeep for supporting this research and providing computational resources.

## References


Aebersold, R. and Mann, M. Mass-spectrometric exploration of proteome structure and function. *Nature*, 537 (7620):347–355, 2016. doi: 10.1038/nature19949.

Albergo, M. S. and Vanden-Eijnden, E. Building normalizing flows with stochastic interpolants. In *International Conference on Learning Representations (ICLR)*, 2023.

Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and van den Berg, R. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pp. 17981–17993, 2021.

Bassani-Sternberg, M., Bräunlein, E., Klar, R., Engleitner, T., Sinitcyn, P., Audehm, S., Straub, M., Weber, J., Slotta-Huspenina, J., Specht, K., Martignoni, M. E., Werner, A., Hein, R., Busch, D. H., Peschel, C., Rad, R., Cox, J., Mann, M., and Krackhardt, A. M. Direct identification of clinically relevant neoepitopes presented on native human melanoma tissue by mass spectrometry. *Nature Communications*, 7:13404, 2016. doi: 10.1038/ncomms13404.

Bonissone, S. R. and Pevzner, P. A. Immunoglobulin classification using the colored antibody graph. *Journal of Computational Biology*, 23(6):483–494, 2016. doi: 10.1089/cmb.2016.0010.

Campbell, A., Yim, J., Barzilay, R., Rainforth, T., and Jaakkola, T. Generative flows on discrete state-spaces: Enabling multimodal flows with applications to protein co-design. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 5296–5325, 2024.

Cao, Z., Qin, T., Liu, T.-Y., Tsai, M.-F., and Li, H. Learning to rank: From pairwise approach to listwise approach. In *Proceedings of the 24th International Conference on Machine Learning (ICML)*, pp. 129–136, 2007. doi: 10.1145/1273496.1273513.

Dancik, V., Addona, T. A., Clauser, K. R., Vath, J. E., and Pevzner, P. A. De novo peptide sequencing via tandem mass spectrometry. *Journal of Computational Biology*, 6 (3-4):327–342, 1999. doi: 10.1089/106652799318300.

Dao, T. FlashAttention-2: Faster attention with better parallelism and work partitioning. In *International Conference on Learning Representations (ICLR)*, 2024.

Eloff, K., Kalogeropoulos, K., Mabona, A., Morell, O., Catzel, R., Rivera-de Torre, E., Jespersen, J. B., Williams, W., van Beljouw, S. P. B., Skwark, M. J., Laustsen, A. H., Brouns, S. J. J., Ljungars, A., Schoof, E. M., Van Goey, J., auf dem Keller, U., Beguir, K., Lopez Carranza, N., and Jenkins, T. P. InstaNovo enables diffusion-powered de novo peptide sequencing in large-scale proteomics experiments. *Nature Machine Intelligence*, 7(4):565–579, 2025. doi: 10.1038/s42256-025-01019-5.

Gat, I., Remez, T., Shaul, N., Kreuk, F., Chen, R. T. Q., Synnaeve, G., Adi, Y., and Lipman, Y. Discrete flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pp. 40149–40183, 2024.

Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.

Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4195–4205, 2023. doi: 10.1109/ICCV51070.2023.00387.

Petrovskiy, D. V., Nikolsky, K. S., Rudnev, V. R., Kulikova, L. I., Butkova, T. V., Malsagova, K. A., Kopylov, A. T., and Kaysheva, A. L. PowerNovo2: A generative flow-based approach to non-autoregressive de novo peptide sequencing. *PLOS Computational Biology*, 22(5): e1014298, 2026. doi: 10.1371/journal.pcbi.1014298.

Roepstorff, P. and Fohlman, J. Proposal for a common nomenclature for sequence ions in mass spectrometry of peptides. *Biomedical Mass Spectrometry*, 11(11):601, 1984. doi: 10.1002/bms.1200111109.

Stark, H., Jing, B., Wang, C., Corso, G., Berger, B., Barzilay, R., and Jaakkola, T. Dirichlet flow matching with applications to DNA sequence design. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46452–46479, 2024.

Tran, N. H., Zhang, X., Xin, L., Shan, B., and Li, M. De novo peptide sequencing by deep learning. *Proceedings of the National Academy of Sciences*, 114(31):8247–8252, 2017. doi: 10.1073/pnas.1705691114.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pp. 5998–6008, 2017.

Wang, M., Wang, J., Carver, J., Pullman, B. S., Cha, S. W., and Bandeira, N. Assembling the community-scale discoverable human proteome. *Cell Systems*, 7(4):412–421.e5, 2018. doi: 10.1016/j.cels.2018.08.004.

Yilmaz, M., Fondrie, W. E., Bittremieux, W., Oh, S., and Noble, W. S. De novo mass spectrometry peptide sequencing with a transformer model. In *Proceedings of*

*the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pp. 25514–25522, 2022.

Yilmaz, M., Fondrie, W. E., Bittremieux, W., Melendez, C. F., Nelson, R., Ananth, V., Oh, S., and Noble, W. S. Sequence-to-sequence translation from mass spectra to peptides with a transformer model. *Nature Communications*, 15(1):6427, 2024. doi: 10.1038/s41467-024-49731-x.

# A. Implementation Details, Hyperparameters, and Hardware

## A.1. Model Architecture Specifications

Table 4 summarizes the comprehensive architectural specifications for DFlowNovo. The entire model comprises 59,496,750 parameters ( 59.50M), partitioned between the Spectrum Transformer Encoder (16.29M, 27.4%), the Discrete Flow Decoder (42.85M, 72.0%), and the Precursor Length Predictor (0.36M, 0.6%).

| Module | Hyperparameter | Value |
|---|---|---|
| **Spectrum Encoder** | Transformer Layers | 6 |
| | Hidden Dimension ($d_{\text{model}}$) | 512 |
| | Attention Heads | 8 |
| | Feed-Forward Dimension ($d_{\text{ffn}}$) | 1536 |
| | Attention Implementation | FlashAttention-2 |
| | Max Input Centroided Peaks ($P$) | 200 |
| | Peak Embeddings | Dual Sinusoidal (Fragment + Complementary) |
| **Discrete Flow Decoder** | Transformer Blocks | 6 |
| | Hidden Dimension ($d_{\text{model}}$) | 512 |
| | Attention Heads | 8 |
| | Feed-Forward Dimension ($d_{\text{ffn}}$) | 1536 |
| | Feed-Forward Activation | SwiGLU |
| | Time & Condition Mechanism | AdaLN-Zero on $(t, M_{\text{prec}}, z, L)$ |
| | Categorical Vocabulary Size ($S$) | 32 (20 Canonical + 10 PTMs + [PAD] + [M]; 30 active classes) |
| **Length Classifier** | Architecture | 2-Layer MLP on $h_{\text{[CLS]}}$ |
| | Hidden Dimension | 512 |
| | Output Classes ($L$) | $1 \leq L \leq 50$ (50 classes) |
| **Inference & Decoding** | Reverse Euler Steps ($K$) | 20 (Cosine Schedule) |
| | Detailed Balance Rate ($\eta$) | 0.15 (Window $t \in [0.2, 0.8]$) |
| | Candidate Pool Size ($C$) | 6 (Hybrid Beam) |
| | Greedy Decoding Temp ($T$) | 0.0 (Candidate 1, MAP argmax) |
| | Stochastic Decoding Temp ($T$) | 0.7 (Candidates 2–6, categorical sampling) |
| | Mass Grid Resolution ($\Delta m$) | 0.02 Da (Tolerance $\tau = \pm 0.1$ Da, $W = 11$) |

*Table 4.* **Comprehensive Hyperparameter Specification for DFlowNovo.**

## A.2. Training Protocol and Optimization

DFlowNovo was trained on the MassIVE-KB repository ($N = 28{,}232{,}112$ human spectra) using PyTorch 2.5 and PyTorch Lightning. Optimization utilized AdamW with $\beta_1 = 0.9, \beta_2 = 0.98, \epsilon = 10^{-8}$, and weight decay of $10^{-4}$. The learning rate followed a Warmup-Stable-Decay (WSD) schedule with warmup ratio 0.05, stable ratio 0.70, peak learning rate $\eta_{\max} = 2 \times 10^{-4}$, and minimum learning rate ratio 0.01. Training utilized mixed-precision (bfloat16) on NVIDIA H100 80GB GPUs with an effective batch size of 1,024 spectra.

## A.3. Code and Data Availability

- **Source Code and Checkpoints:** The official implementation of DFlowNovo, training configurations, inference scripts, precomputed knapsack tables, and evaluation pipelines are openly available at https://github.com/joelinator/JoelResearch. Pretrained model weights are hosted on the Hugging Face Hub at https://huggingface.co/joelinator/dflow-novo-model.
- **MassIVE-KB Pretraining Corpus:** Tandem mass spectrometry spectra from the human MassIVE-KB dataset are publicly hosted by UCSD under MassIVE accession MSV000084075 (https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?accession=MSV000084075).
- **Nine-Species Benchmark Dataset:** The multi-species tandem mass spectrometry benchmark dataset ($N = 2{,}322{,}730$ spectra across nine organisms) is available via the PRIDE repository under accession PXD004948 and hosted on Hugging Face Datasets at https://huggingface.co/datasets/InstaDeepAI/ms_ninespecies_benchmark.

## B. CTMC Discrete Flow Matching Mathematical Foundations

In this section, we provide the concise mathematical foundations for Continuous-Time Markov Chain (CTMC) Discrete Flow Matching on probability simplices, formalizing the framework of Campbell et al. (Campbell et al., 2024).

### B.1. Continuous-Time Master Equation and Rate Matrices

Let $\mathcal{X} = \{1, \ldots, S\}$ be a finite categorical state space of size $S = 21$ (20 amino acids + $[M]$). The time evolution of the marginal probability vector $\mathbf{p}_t \in \Delta^{S-1}$ under a continuous-time Markov chain is governed by:

$$\frac{d}{dt}\mathbf{p}_t = \mathbf{p}_t R_t \tag{17}$$

where $R_t \in \mathbb{R}^{S\times S}$ is the transition rate matrix satisfying $R_t(j \mid i) \geq 0$ for $i \neq j$, and diagonal elements $R_t(i \mid i) = -\sum_{j\neq i} R_t(j \mid i)$.

### B.2. Conditional Jump-Minimal Dynamics

Given target token $y \in \mathcal{V}_{\text{AA}}$ at $t = 1$ and initial state $[M]$ at $t = 0$, the masked conditional probability path is:

$$p_{t|1}(x \mid y) = \big(1 - \kappa(t)\big)\delta_{[M]}(x) + \kappa(t)\delta_y(x) \tag{18}$$

where $\kappa(t)$ is a smooth schedule with $\kappa(0) = 0, \kappa(1) = 1$. Differentiating with respect to $t$:

$$\frac{d}{dt}p_{t|1}(x \mid y) = -\dot{\kappa}(t)\delta_{[M]}(x) + \dot{\kappa}(t)\delta_y(x) \tag{19}$$

Under jump-minimal transition dynamics, transitions occur directly from $[M]$ to $y$:

$$R_t(y \mid [M], y) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \tag{20}$$

For any other pair $(i, j)$ with $i \neq [M]$ or $j \neq y$, $R_t(j \mid i, y) = 0$.

### B.3. Simulation-Free Marginal Vector Field

Marginalizing over target sequences $\mathbf{y} \sim q(\mathbf{y} \mid \mathcal{S})$, the marginal transition rate from mask $[M]$ to residue $a \in \mathcal{V}_{\text{AA}}$ is:

$$R_t(a \mid [M], \mathcal{S}) = \sum_y R_t(a \mid [M], y) q(y \mid x = [M], \mathcal{S}) \tag{21}$$

$$= \frac{\dot{\kappa}(t)}{1 - \kappa(t)}\hat{p}_{1|t}(a \mid \mathbf{x}_t, \mathcal{S}) \tag{22}$$

where $\hat{p}_{1|t}(a \mid \mathbf{x}_t, \mathcal{S})$ is the neural posterior predicted by the discrete flow decoder. Training by minimizing cross-entropy over masked positions (Equation (10)) directly minimizes the KL divergence between the true velocity field and the parameterized model.

## C. Combinatorial KnapsackDP Exactness and Complexity

### C.1. Formulation and Exactness

Let $\mathcal{W} = \{\lfloor m(a)/\Delta m \rceil \mid a \in \mathcal{V}_{\text{AA}}\} \subset \mathbb{N}$ denote the set of integer-quantized amino acid masses on grid $\Delta m = 0.02\,\text{Da}$. Let $T[k, b] \in \{0, 1\}$ be the boolean dynamic programming table defined in Equation (11).

**Proposition C.1** (Reachability Exactness)**.** *For any positive integer $k \leq K_{\max}$ and mass bin $b \leq B_{\max}$, $T[k, b] = 1$ if and only if there exists a multiset of amino acids $\{a_1, \ldots, a_k\} \subseteq \mathcal{V}_{AA}$ such that $\sum_{i=1}^{k} \lfloor m(a_i)/\Delta m \rceil = b$.*

*Proof.* By mathematical induction on residue count $k$. For $k = 0$, $T[0, 0] = 1$ and $T[0, b] = 0$ for all $b > 0$, which holds trivially. Assume the proposition holds for $k - 1$. For step $k$:

$$T[k, b] = \bigvee_{a\in\mathcal{V}_{\text{AA}}} T[k - 1, b - w_a] \tag{23}$$

**Algorithm 2** Precomputed GPU `KnapsackDP` Table Construction

**Input:** Residue masses $\{m(a)\}_{a\in\mathcal{V}_{\text{AA}}}$, grid resolution $\Delta m = 0.02\,\text{Da}$, max mass $M_{\max} = 5000\,\text{Da}$, max length $K_{\max} = 50$, instrument tolerance $\tau = 0.1\,\text{Da}$.
Initialize bin count $B_{\max} \leftarrow \lfloor M_{\max}/\Delta m \rfloor = 250{,}000$.
Allocate boolean table $T \in \{0,1\}^{(K_{\max}+1)\times B_{\max}} \leftarrow 0$; set $T[0,0] \leftarrow 1$.
**for** $k = 1$ **to** $K_{\max}$ **do**
  **for** $a \in \mathcal{V}_{\text{AA}}$ **do**
    $w_a \leftarrow \lfloor m(a)/\Delta m \rceil$
    $T[k, w_a : B_{\max}] \leftarrow T[k, w_a : B_{\max}] \vee T[k-1, 0 : B_{\max} - w_a]$
  **end for**
**end for**
Window size $W \leftarrow 2\lfloor \tau/\Delta m \rfloor + 1 = 11$.
Apply GPU 1D max-pooling with kernel size $W$:
$T_{\text{pooled}}[k,b] \leftarrow \max_{|j-b|\le 5} T[k,j]$.
**Output:** Pooled reachability tensor $T_{\text{pooled}}$ in GPU memory.

$T[k,b] = 1 \iff \exists a \in \mathcal{V}_{\text{AA}}$ such that $T[k-1, b-w_a] = 1$. By the inductive hypothesis, $T[k-1,b-w_a] = 1 \iff \exists\{a_1,\ldots,a_{k-1}\}$ such that $\sum_{i=1}^{k-1} w_{a_i} = b - w_a$. Setting $a_k = a$ yields $\sum_{i=1}^{k} w_{a_i} = b$. Thus, $T[k,b] = 1$ if and only if the mass bin $b$ is reachable by exactly $k$ residues. □

### C.2. Computational Complexity

- **Precomputation:** Computing table $T$ requires $K_{\max} \times |\mathcal{V}_{\text{AA}}| \times B_{\max} = 50 \times 20 \times 250{,}000 = 2.5 \times 10^8$ boolean operations, executing in under $0.25$ seconds on GPU once at initialization.
- **Inference Lookup:** At each flow step $k$, checking chemical admissibility (Equation (13)) consists of a single memory indexing operation $T_{\text{pooled}}[u-1, b]$ per residue candidate. This requires $\mathcal{O}(1)$ time per residue, introducing negligible overhead ($< 1.8$ ms per batch of 512 spectra).

## D. Exhaustive Multi-Species Cross-Kingdom Generalization

To evaluate whether the learned continuous-time vector field generalizes across phylogenetic diversity without species-level overfitting, Table 5 reports the complete evaluation of DFlowNovo across all 2,322,730 spectra of the Nine-Species dataset. Pretrained exclusively on human MassIVE-KB spectra, non-human species represent zero-shot cross-organism evaluation.

| Species Display Name | Taxonomic Group | Spectrum Count | Peptide Acc. (%) | Mass Match (%) | AA F1 (%) | AA Precision (%) | AA Recall (%) | Length Acc. (%) |
|---|---|---|---|---|---|---|---|---|
| *Mus musculus* (Mouse) | Mammalia | 25,963 | 48.06 | 51.94 | 75.02 | 75.35 | 74.69 | 76.86 |
| *Homo sapiens* (Human) | Mammalia | 43,416 | 65.14 | 67.86 | 78.60 | 78.76 | 78.44 | 84.98 |
| *Candidatus endoloripes* | Bacteria | 79,144 | 48.98 | 52.57 | 71.18 | 71.46 | 70.91 | 75.09 |
| *Vigna mungo* (Black gram) | Plantae | 107,955 | 69.15 | 73.47 | 86.96 | 87.06 | 86.86 | 88.08 |
| *Saccharomyces cerevisiae* (Yeast) | Fungi | 104,163 | 66.44 | 68.24 | 81.16 | 81.28 | 81.04 | 84.77 |
| *Solanum lycopersicum* (Tomato) | Plantae | 177,721 | 69.13 | 71.94 | 81.93 | 82.04 | 81.82 | 86.81 |
| *Apis mellifera* (Honeybee) | Insecta | 192,397 | 55.01 | 58.75 | 72.12 | 72.31 | 71.93 | 78.69 |
| *Methanosarcina mazei* | Archaea | 262,739 | 62.59 | 66.57 | 78.40 | 78.47 | 78.33 | 83.04 |
| *Bacillus subtilis* | Bacteria | 1,329,232 | 65.83 | 68.78 | 77.72 | 77.89 | 77.56 | 84.29 |
| **Macro-Average / Total** | **6 Biological Kingdoms** | **2,322,730** | **61.15** | **64.46** | **78.12** | **78.29** | **77.95** | **82.25** |

*Table 5.* **Comprehensive Per-Species Extended Performance on all Nine Species ($N = 2{,}322{,}730$ spectra total).** Across all nine organisms, DFlowNovo recovers 224,249 unique peptides with a macro-average Amino Acid F1 of 78.12% and Peptide Accuracy of 61.15%.